

《Automotive Configuration》

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Class period: 1-8 week , 2nd term

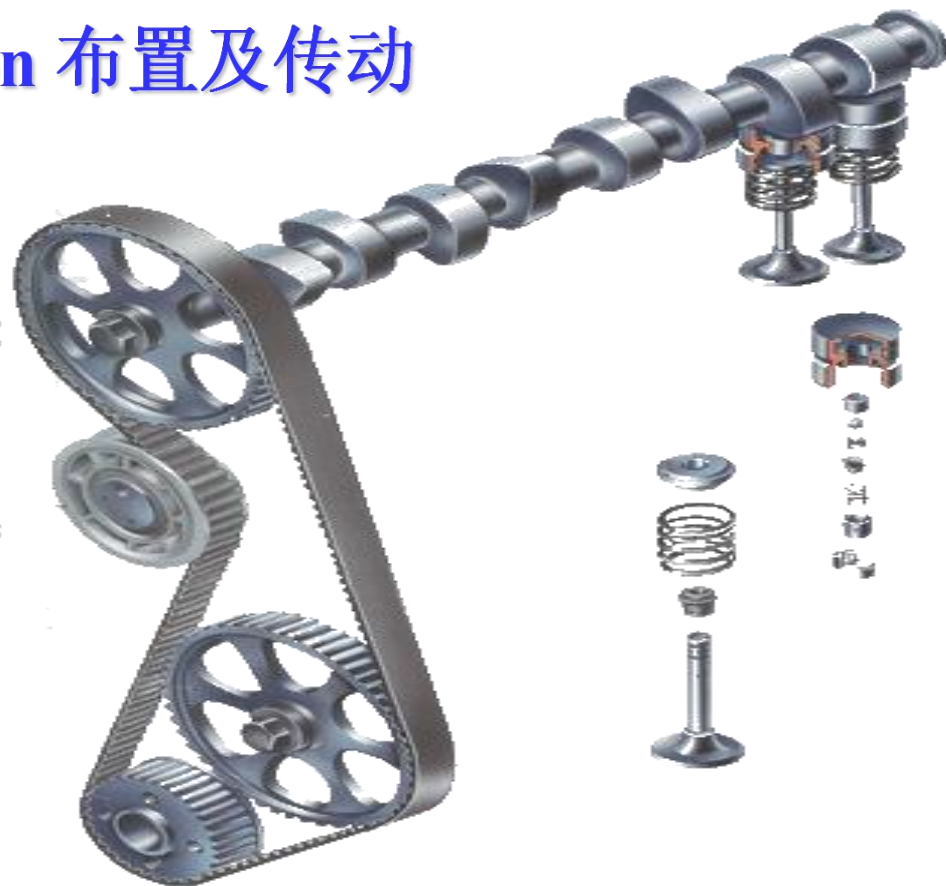
Chapter 3 Valve Trains 配气机构

3.1 Overview 概述

3.2 Layout and Transmission 布置及传动

3.3 Valve Timing 配气正时

3.4 Parts and Components 零件及部件



3.1 Overview 概述

1. Volumetric Efficiency (ϕ_c) 充气效率:

An index for evaluating gas exchange process.
评价气缸内换气过程好坏的指标。

$$\phi_c = \frac{M}{M_0}$$

where M = the actual air intake mass (kg); M_0 = theoretical mass that could be contained in the engine cylinder at ambient conditions in naturally-aspirated engines.

其中, M = 每循环实际进气量(kg); M_0 = 大气环境下 (自然吸气) 充满气缸工作容积的理论质量(kg)。

1. Volumetric Efficiency 充气效率

• $\phi_c \uparrow \longrightarrow$ Actual air inflow (mixture or fuel mass) 实际进气量

(混合气或油量) $\uparrow \longrightarrow$ Heat release 放热量 $\uparrow \longrightarrow$

Total output power 输出总功率 $\uparrow$

In general, $\phi_c = 80\% \sim 90\%$



1. Volumetric Efficiency 充气效率

Influencing Factor: (when fixed volume)

影响因素：（容积一定）

1) Pressure at the end of intake process (intake resistance)

进气终了压力（进气阻力损失）

2) Temperature at the end of intake process (residual gas & high temperature components heating)

进气终了温度（残余废气及高温组件加热）

3) Residual gas existence 残余废气存在



2. Constitution of Valve Trains 配气机构的组成

视频 (Video)

- 1) Valve transmission group 气门传动组
- 2) Valve group 气门组

1) Valve transmission group consists of all parts that begin from timing gear to the part pushing valves action.

气门传动组是从正时齿轮开始至推动气门动作的所有组件。

1) Valve transmission group 气门传动组

1-Camshaft timing gear

凸轮轴正时齿轮

2-Camshaft

凸轮轴

3-Tappet

挺柱

4-Push rod

推杆

5-Rocker arm shaft bracket

摇臂轴支架；

6-Rocher arm shaft 摇臂轴；

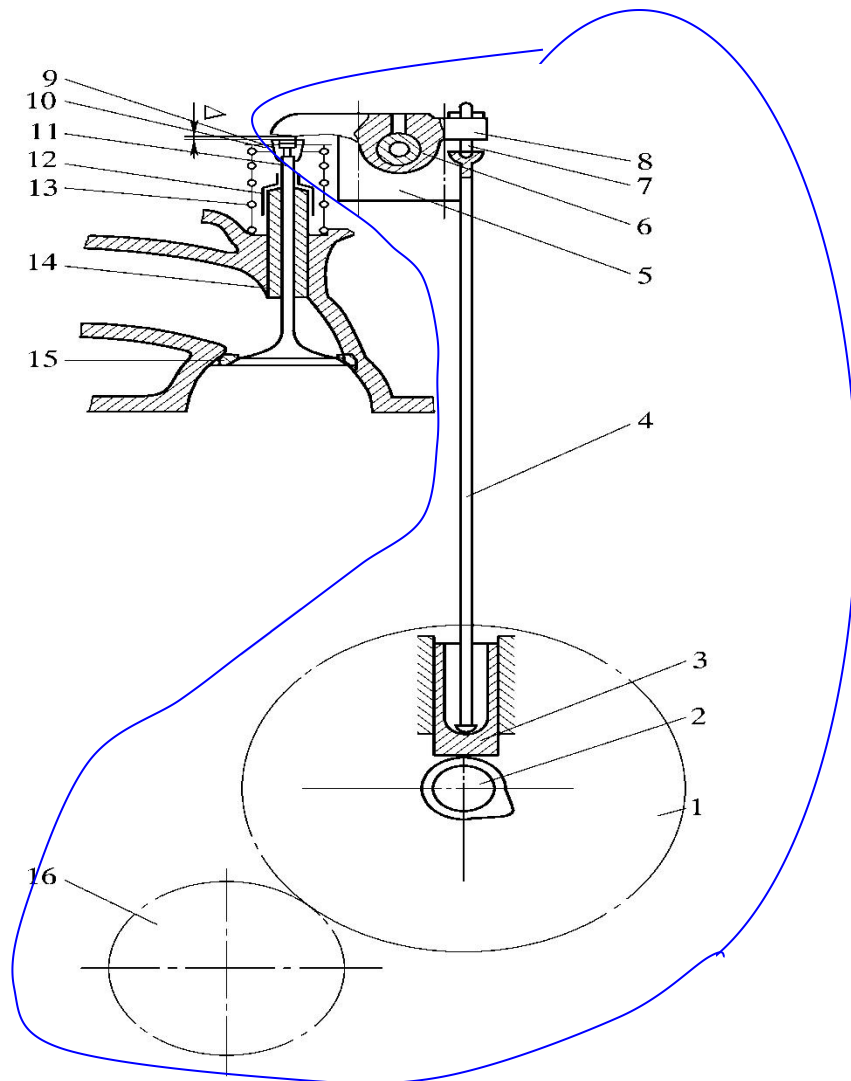
7-Adjust screw and lock nut

调整螺钉及锁紧螺母；

8-Rocker arm 摇臂；

16-Crankshaft timing gear

曲轴正时齿轮；



2) Valve group 气门组:

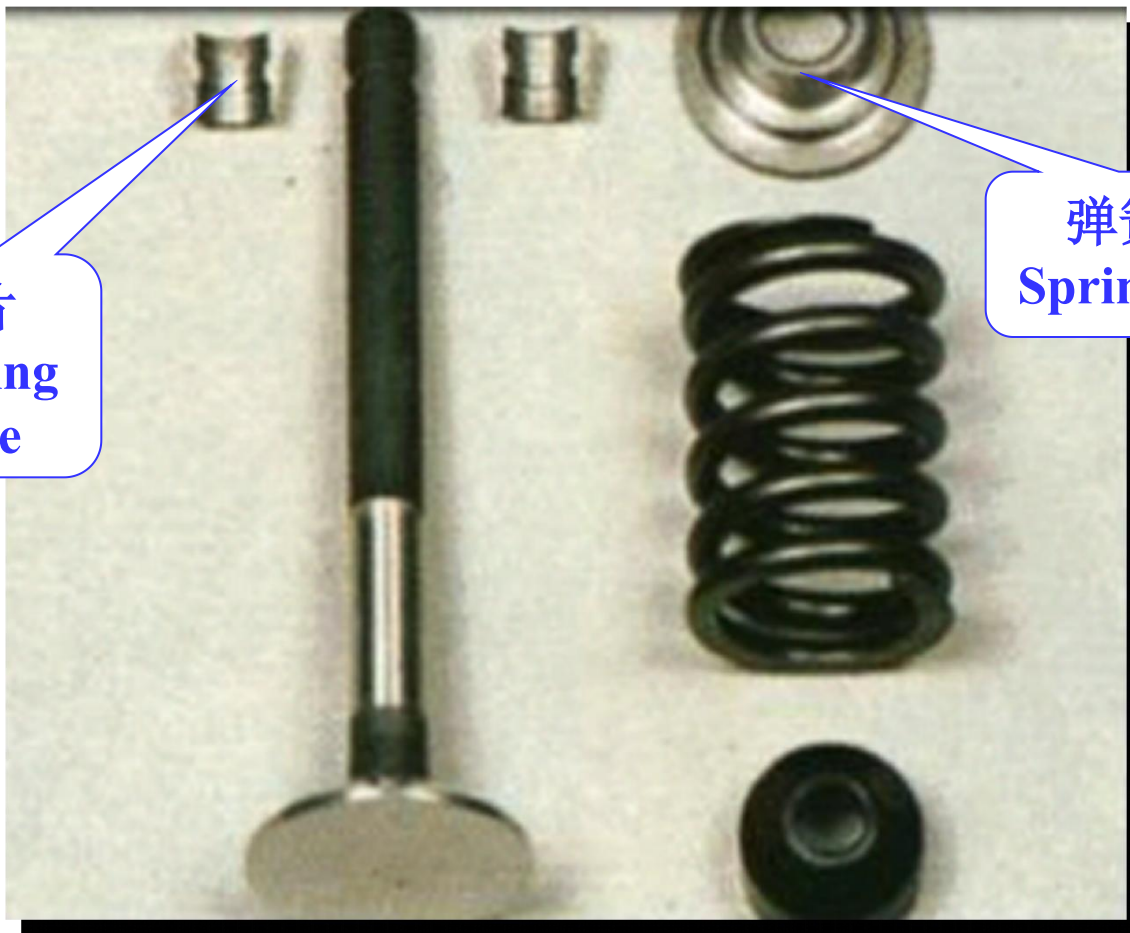
Its function is mainly for keeping valves closing.

其功用主要是维持气门的关闭。



Valve group Physical Map 气门组实物图

锁片
Locking
plate

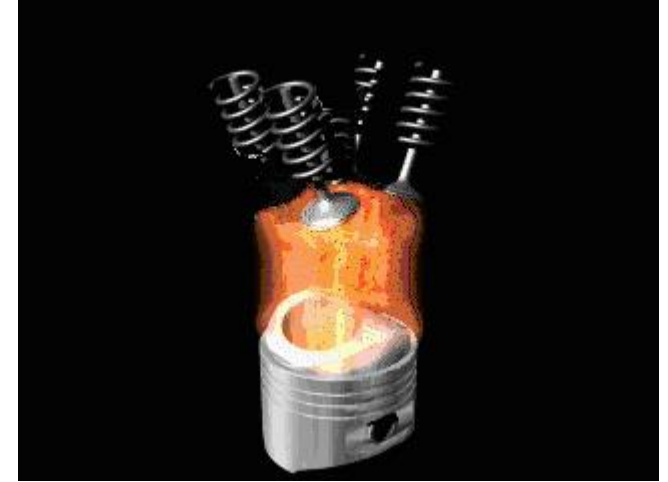


弹簧座
Spring seat



3. Valve train function 配气机构功用

Valve train opens and closes intake and exhaust valves at regular time according to the working cycle and firing order in each cylinder, promoting fresh mixture (gasoline engine) or air (diesel engine) to enter the cylinder, and residual gas to exhaust from the cylinder promptly.



按照发动机每个气缸内所进行的工作循环和发火次序的要求，定时开启和关闭气缸的进、排气门，使新鲜可燃混合气(汽油机)或空气(柴油机)得以及时进入气缸，废气得以及时从气缸排出。

4. Demand 要求

- 1) High enough volumetric efficiency 足够高的充气效率
- 2) High reliability 工作可靠
- 3) Accurate valve timing 配气相位准确
- 4) Good high speed dynamic characteristics (without drive chain flying out, bouncing or early seating)

气门机构有良好的高速动力特性; (传动链不飞脱、不反跳、不提前落座)

3.2 Layout and Transmission 布置与传动

1. Valve layout type

气门布置形式



气门侧置
Side-mounted
Valve



气门顶置
Overhead Valve

2. Camshaft layout type 凸轮轴布置形式



凸轮轴下置
Bottom-mounted
Camshaft

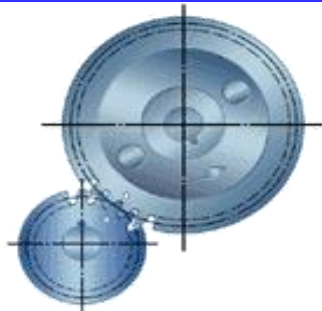


凸轮轴中置
Center-mounted
Camshaft



凸轮轴顶置
Overhead
Camshaft

3. Camshaft transmission type 凸轮轴传动方式

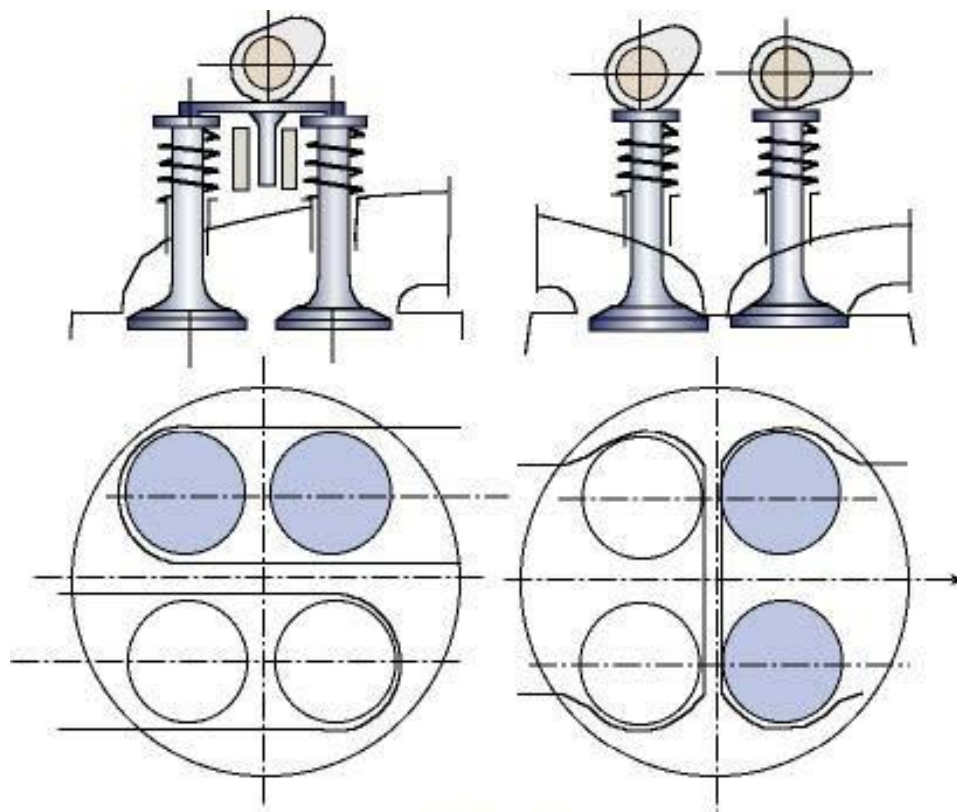
Type	Transmission Route 传动路线	Graphic图示	Application 应用
Gear Drive 齿轮传动	Crankshaft timing gear 曲轴正时齿轮 (steel 钢)→Camshaft timing gear (cast iron 铸铁 or bakelite 胶木)		Center & bottom mounted camshaft valve train 中、下置凸轮轴配气机构
Chain Drive 链条传动	Crankshaft 曲轴→ Chain 链条→ Camshaft timing gear 凸轮轴正时齿轮		Overhead camshaft valve train 顶置凸轮轴配气机构
Toothed Belt Drive 齿形带传动	Crankshaft 曲轴 →Tooth belt 齿形带 →Camshaft timing gear 凸轮轴正时齿轮		Overhead camshaft valve train (lower noise, mass & cost than chain drive 相对链条更低成本、质量及噪声)

4. Valve number and arrangement 气门数目及排列方式

By valve number in each cylinder:

按每缸气门数目分:

2/3/4/5 valve type



四气门式

4 Valve type

5. Valve Clearance 气门间隙:

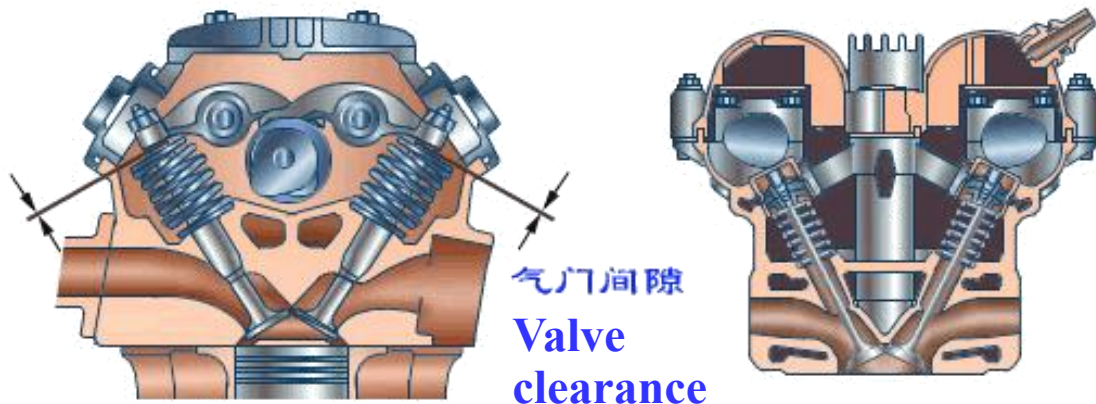
In order to avoid valve expansion and blow-by gas because of high temperature,

Exhaust valve clearance > Intake valve clearance, Why?

为何排气门间隙 > 进气门间隙?

为保证气门关闭严密，避免漏气，在装配时，在气门杆尾端与气门驱动零件（摇臂、挺柱或凸轮）之间留有适当的间隙。

Valve 气门	Clearance 间隙
Intake 进气	0.25~0.30mm
Exhaust 排气	0.30~0.35mm



3.3 Valve Timing 配气相位

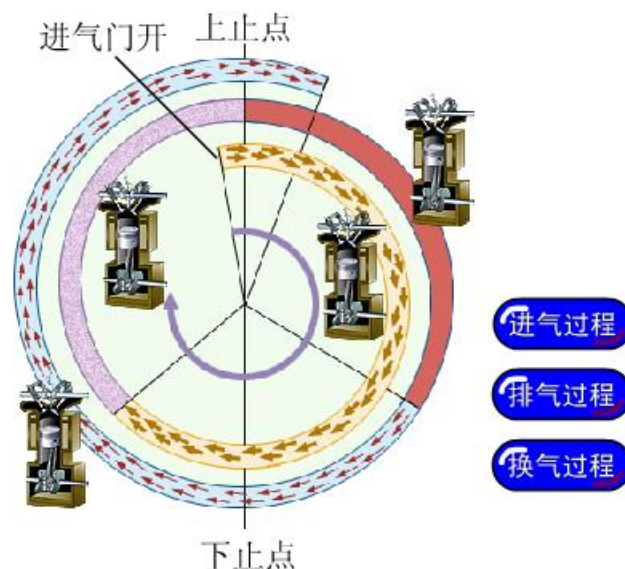
1. Valve timing is the precise crank angle of the opening and closing of the valves.

用曲轴转角表示的进、排气门开闭时刻和开启持续时间，称为配气相位（相位）。

视频 (Video)

Ordinarily, circular diagram is used to show valve timing.

通常用环形图表示配气相位图

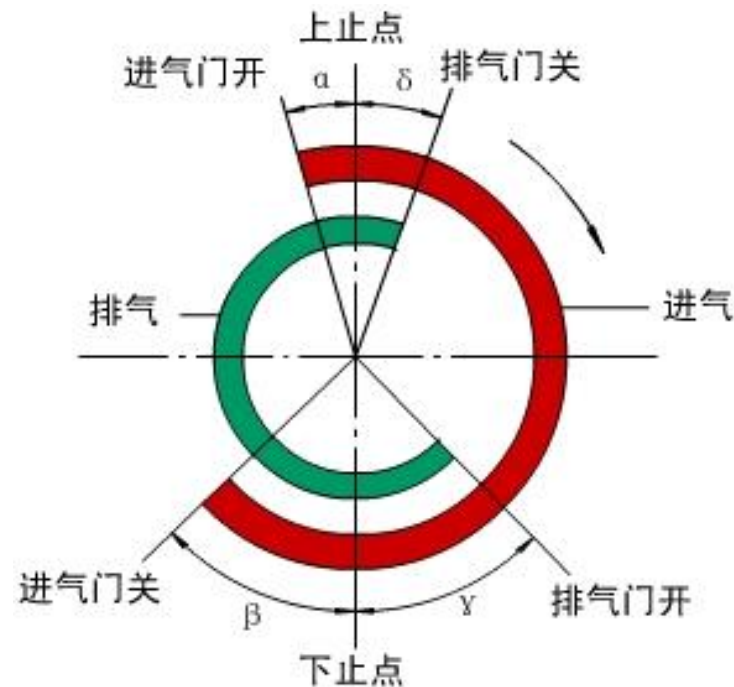


2. Intake Valve Timing 进气门配气相位

1) Intake advance angle (α) 进气提前角:

α is the crank angle corresponding to the duration from intake valve opening to top dead center. In general, α is $10^\circ \sim 30^\circ$.

在排气冲程接近终了，活塞到达上止点之前，进气门便开始开启。从进气门开始开启到上止点所对应的曲轴转角称为进气提前角(或早开角)。 α 一般为 $10^\circ \sim 30^\circ$

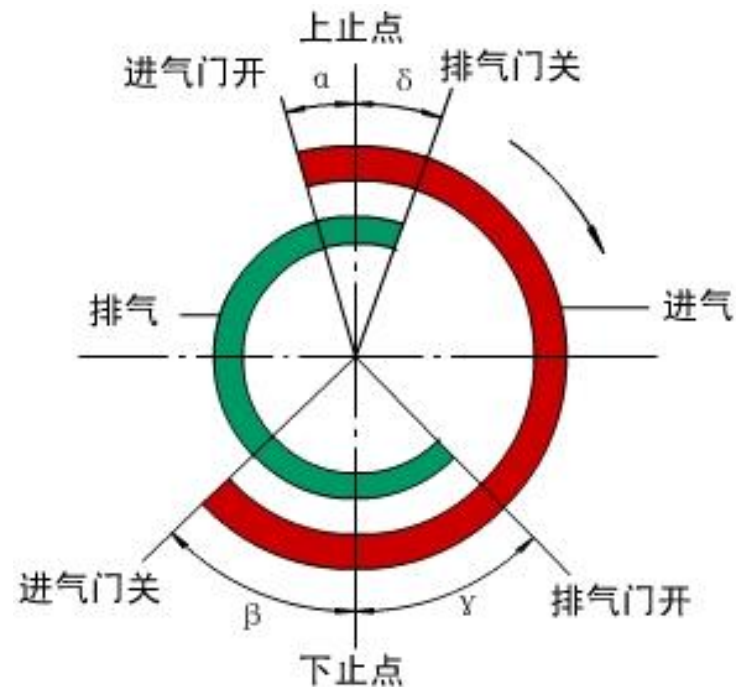


2. Intake Valve Timing 进气门配气相位

2) Intake lag angle (β) 进气迟关角:

β is the crank angle corresponding to the period from bottom dead center to intake valve closing. In general, β is $40^\circ \sim 80^\circ$.

在进气冲程下止点过后，活塞继续上行一段，进气门才关闭。从下止点到进气门关闭所对应的曲轴转角称为进气迟后角(或晚关角)。 β 一般为 $40^\circ \sim 80^\circ$ 。

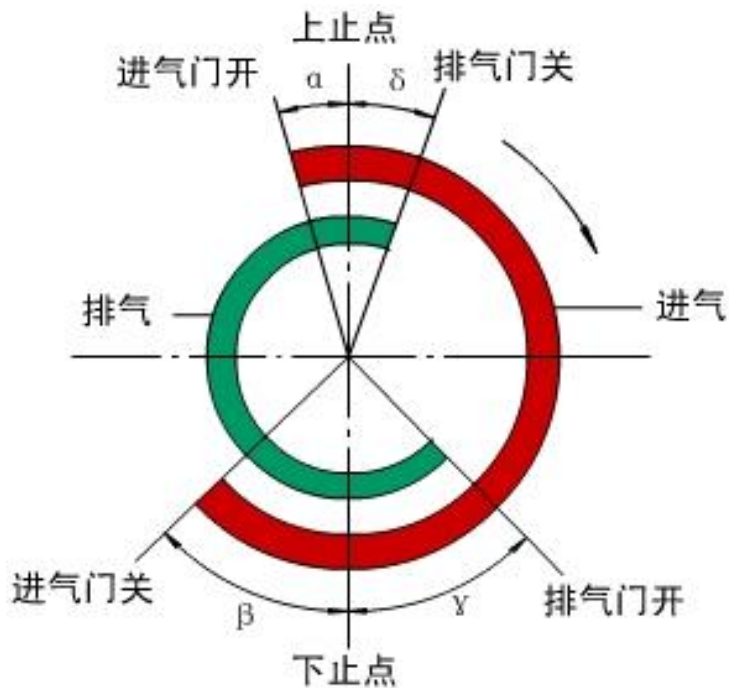


3. Exhaust Valve Timing 排气门配气相位

1) Exhaust advance angle(γ) 排气提前角:

γ is the crank angle corresponding to the duration from exhaust valve opening to bottom dead center. In general, γ is $40^\circ \sim 80^\circ$.

在作功行程的后期，活塞到达下止点前，排气门便开始开启。从排气门开始开启到下止点所对应的曲轴转角称为排气提前角(或早开角)。排气提前角用 γ 表示， γ 一般为 $40^\circ \sim 80^\circ$ 。

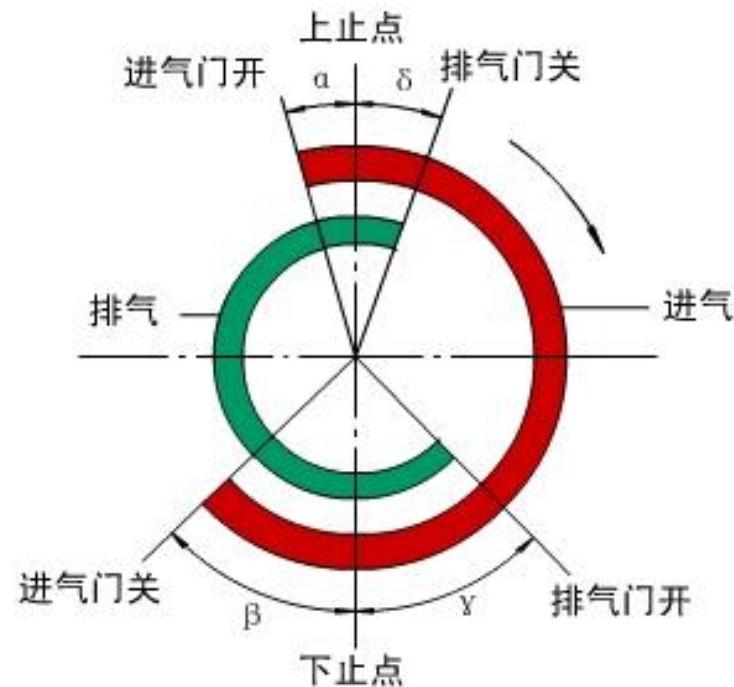


3. Exhaust Valve Timing 排气门配气相位

2) Exhaust lag angle (δ) 排气迟关角:

δ is the crank angle corresponding to the duration from top dead center to exhaust valve closing. In general, δ is $10^\circ \sim 30^\circ$.

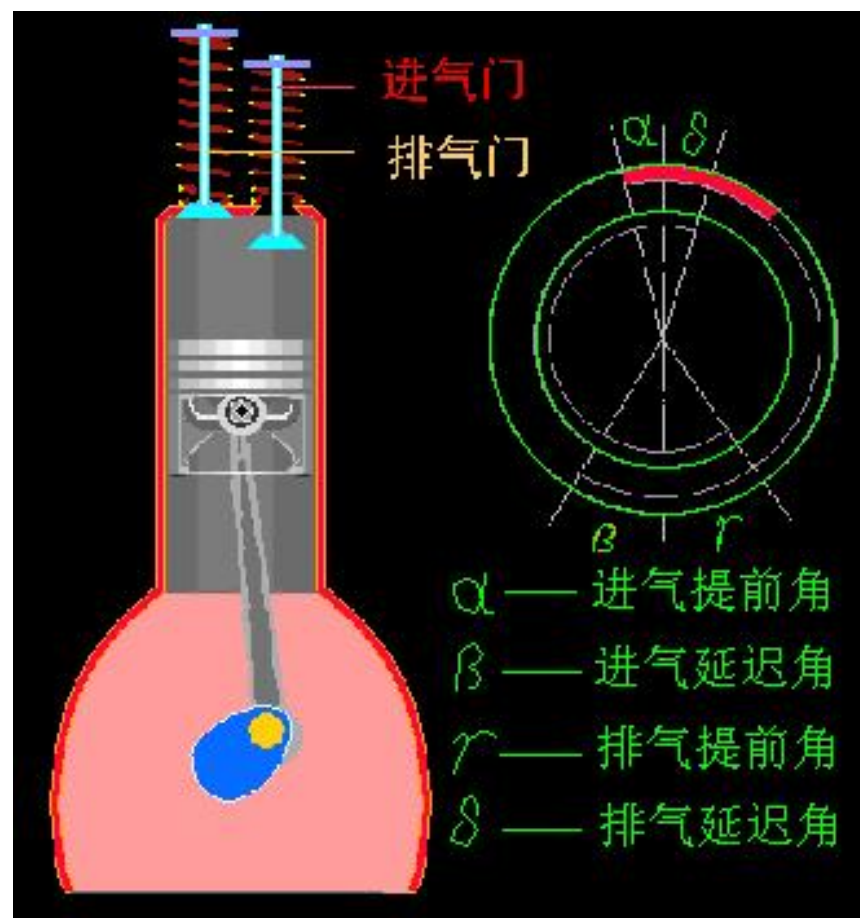
在活塞越过上止点后，排气门才关闭。从上止点到排气门关闭所对应的曲轴转角称为排气迟后角(或晚关角)。 δ 一般为 $10^\circ \sim 30^\circ$ 。



4. Valve Overlap 气门叠开

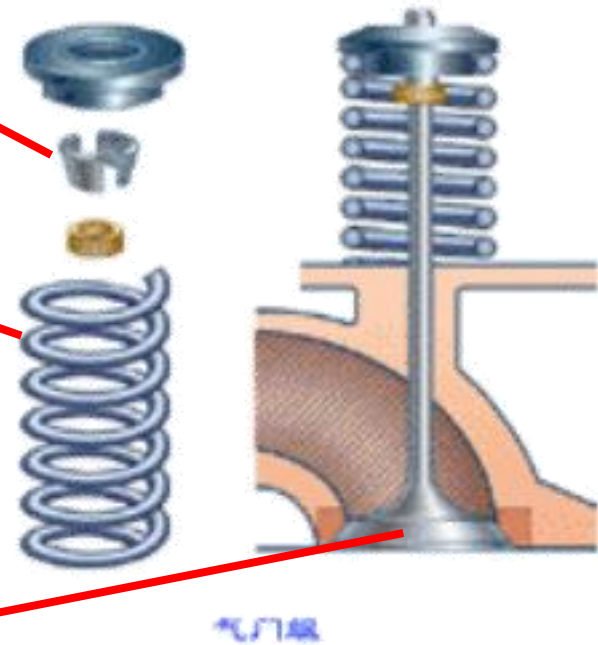
Valve overlap is the phenomenon that both intake and exhaust valves open due to early intake valve opening and late exhaust valve closing. **The valve overlap angle** is the crank angle corresponding to the duration that all valves keep open, namely $\alpha + \delta$.

由于进气门早开和排气门晚关，就出现了一段进排气门同时开启的现象，称为**气门叠开**。同时开启的角度，即进气门早开角与排气门晚关角的和($\alpha + \delta$)，称为**气门叠开角**。



3.4 Components and Parts 配气机构零部件

1. Valve Group 气门组



1. Valve group 气门组

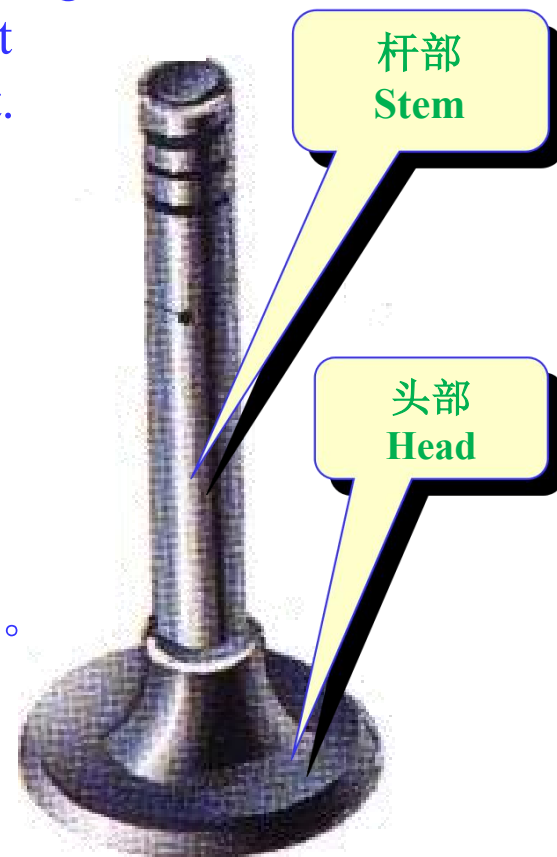
1) Valve 气门

Function: Component parts of combustion chamber; switch for gas intake and exhaust through combustion chamber passage; bearing impact force, high temperature impact and high speed airflow impact.

功用: 燃烧室的组成部分，是气体进出燃烧室的开关，承受冲击力、高温冲击、高速气流冲击。

Working Condition 工作条件:

- A. Intake valve 进气门 570K~670K; Exhaust valve 排气门 1050K~1200K.
- B. Bearing gas pressure, valve spring force, etc. in the valve head. 头部承受气体压力、气门弹簧力等。
- C. Bad cooling and lubrication condition. 冷却和润滑条件差。
- D. Corroded by in-cylinder combustion products. 被缸内燃烧生成物中的物质所腐蚀。



1) Valve 气门 性质及材料视频(Video) 传热及磨损视频(Video)

Property 性质:

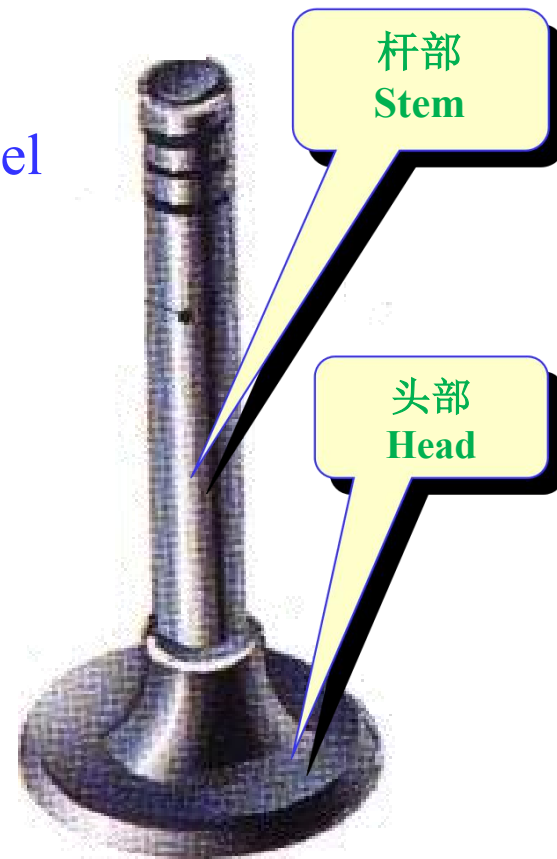
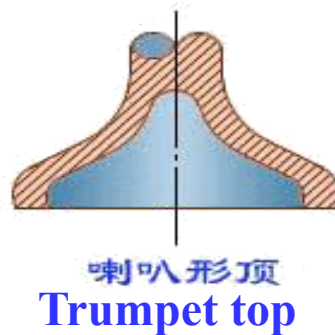
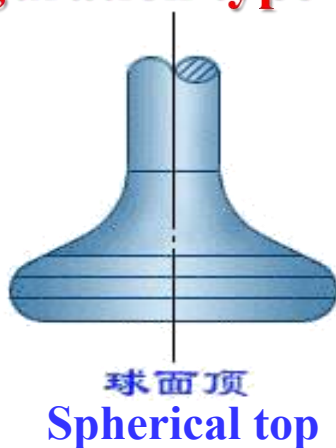
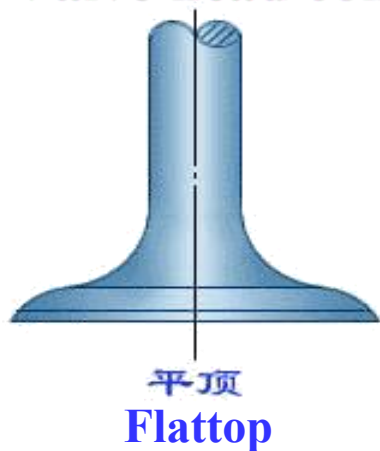
High strength and rigidity 强度、刚度大; heat-resistant, corrosion-resistant, abrasion-resistant. 耐热, 耐腐蚀, 耐磨。

Material 材料:

Intake valve: chrome steel 铬钢 or nickel chromium steel 铬镍钢

Exhaust valve: chrome silicon steel 硅铬钢

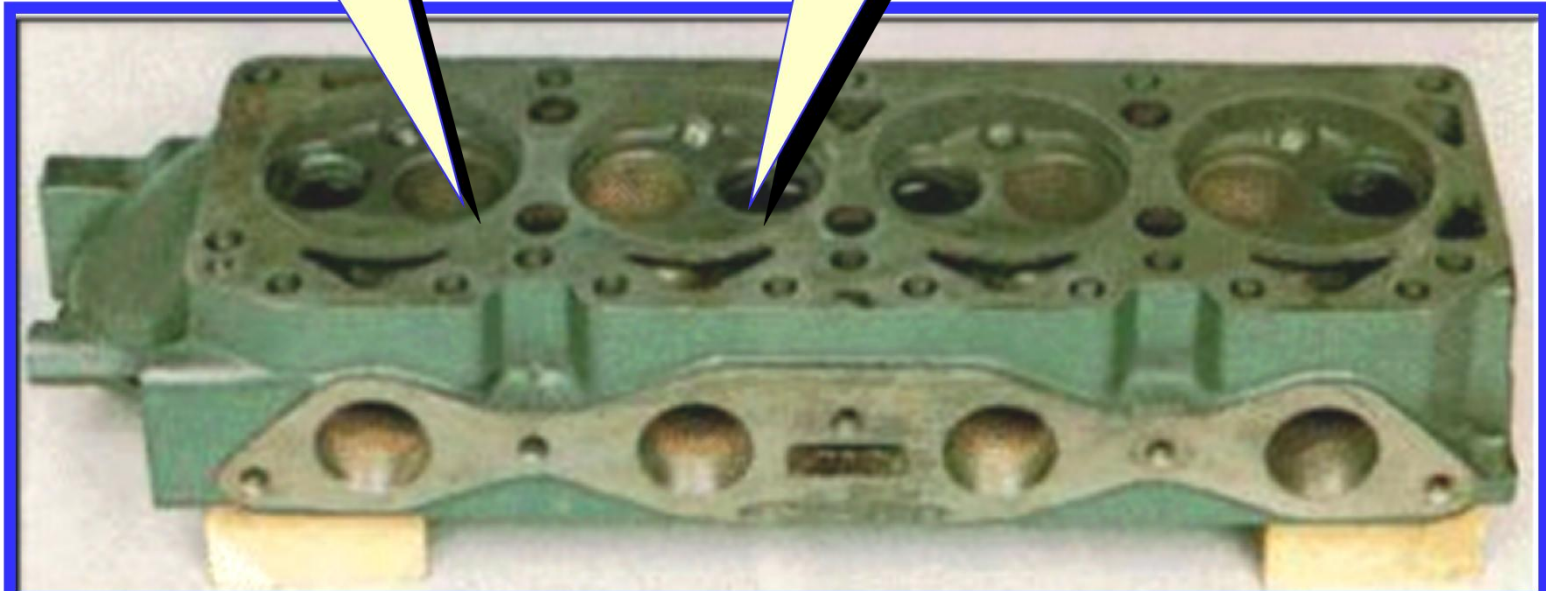
Valve head configuration type 气门头部结构形式:



1) Valve physical map 气门实物图

进气门（大）
Intake valve (bigger)

排气门（小）
Exhaust valve(smaller)



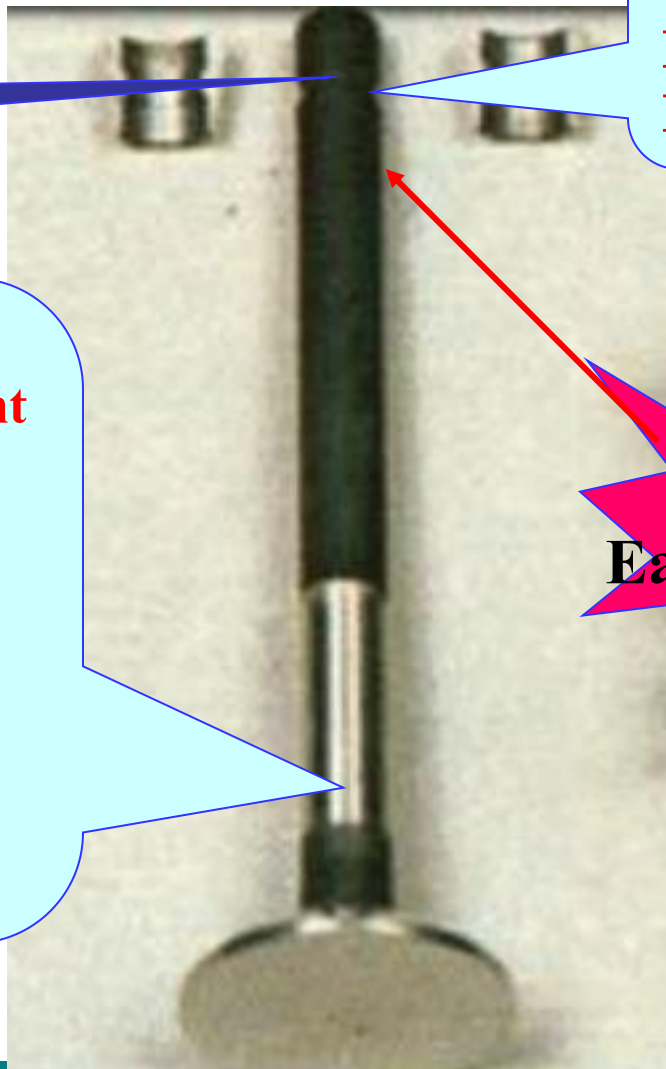
Valve stem 气门杆部

Groove
凹槽

High machining accuracy, heat treatment and burnishing on surface, ensuring high precision and abrasion resistance between the same valve 较高的加工精度，表面经过热处理和磨光，保证同气门导管的配合精度和耐磨性

Valve stem tail 气门杆尾部：
Ring groove 环形槽、
Lockpin hole 锁销孔

易断裂处
Easy to break



2) Valve guide 气门导管

Function: Guide for valve moving, ensuring valve linear motion; heat conduction meanwhile.

作用: 为气门的运动导向, 保证气门直线运动兼起导热作用。

Working conditions: High T, about 500K; difficult lubrication and easy to wear.

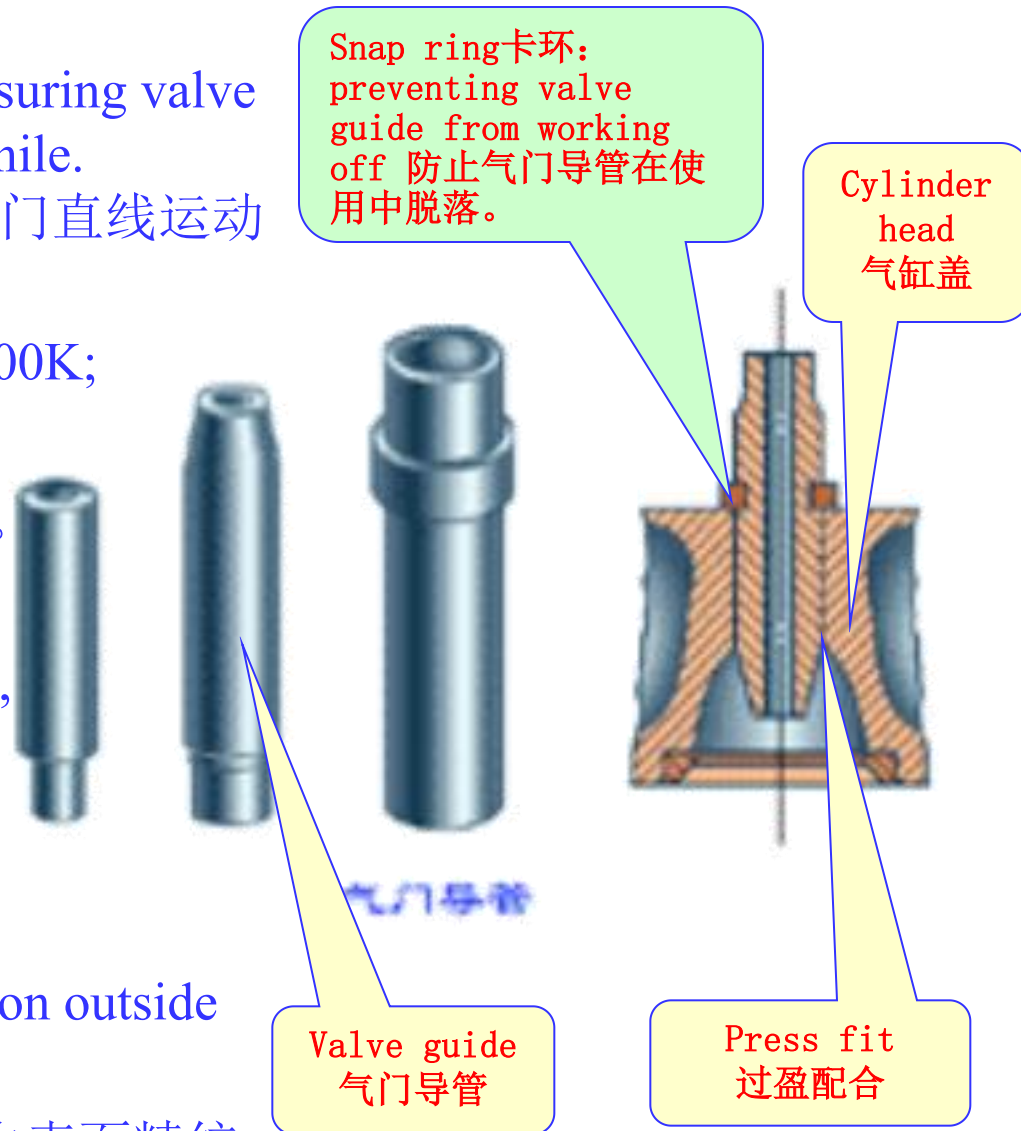
工作条件: 工作温度较高, 约500K。润滑困难, 易磨损。

Material: Cast iron with high graphite, improving self-lubricating.

材料: 用含石墨较多的铸铁, 能提高自润滑作用。

Processing technique: high precision on outside surface, fine twisting on inside surface

加工方法: 外表面加工精度较高, 内表面精绞



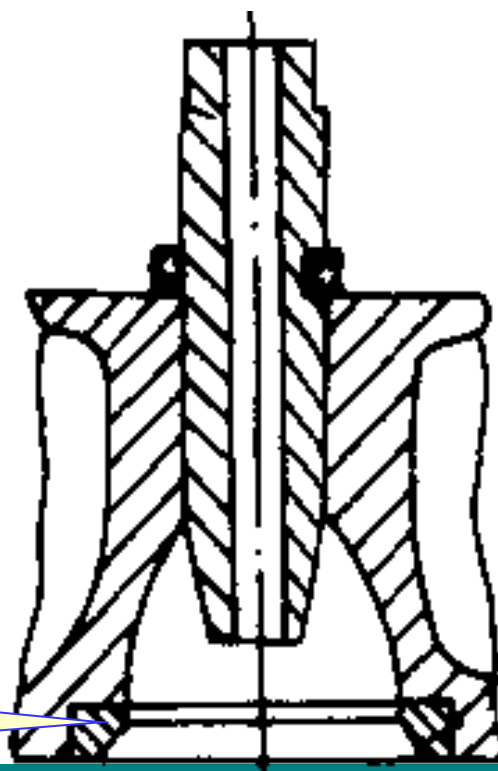
3) Valve seat 气门座

The position where inlet and outlet of cylinder head fit valve conical face.

气缸盖的进、排气道与气门锥面相结合的部位。

Function: Sealing cylinder by tight fitting between its internal cone surface and valve conical face.

作用: 靠其内锥面与气门锥面的紧密贴合密封气缸，接受气门传来的热量。



气门座
Valve seat

Valve seat ring 气门座圈

The ring inlaid in the valve seat ring with big shrink range.

以较大过盈量镶嵌在气门座上的圆环。

Characteristics of Mosaic type valve seat 镶嵌式气门座特点:

Advantage 优点: increasing life length of valve seat, easy to replace.
提高气门座的使用寿命，便于更换。

Disadvantage 缺点: poor thermal conductivity, high precision,
easily resulting in serious accident when working off.
导热性差，加工精度高，脱落时易造成严重事故。

Gasoline engine 汽油机: Exhaust valves apply Mosaic type valve seat.
排气门采用镶嵌式气门座。

Diesel engine 柴油机: Intake valves apply Mosaic type valve seat.
进气门采用镶嵌式气门座。

Valve seat
ring
气门座圈

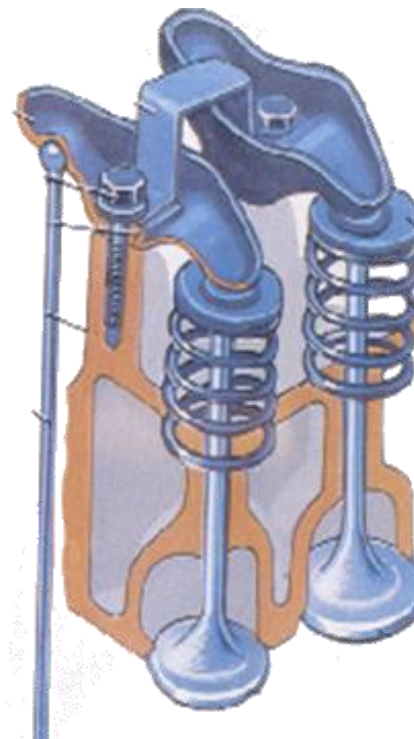
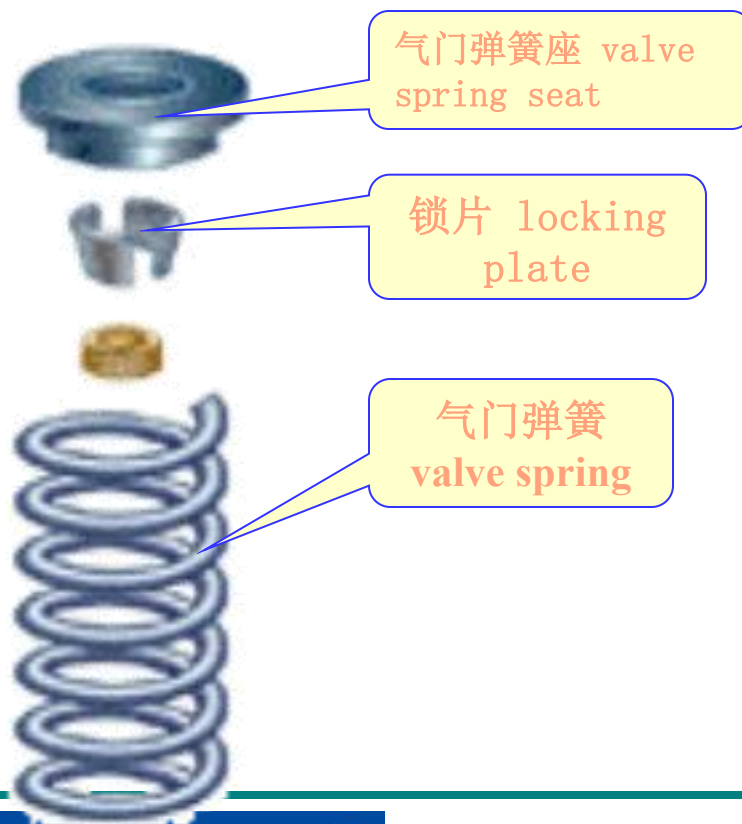


4) Valve spring 气门弹簧 [视频\(Video\)](#)

Function: ensuring valve seating.

功用: 保证气门的回位。

Material: high manganese steel 高锰碳钢
chrome vanadium steel 铬钒钢

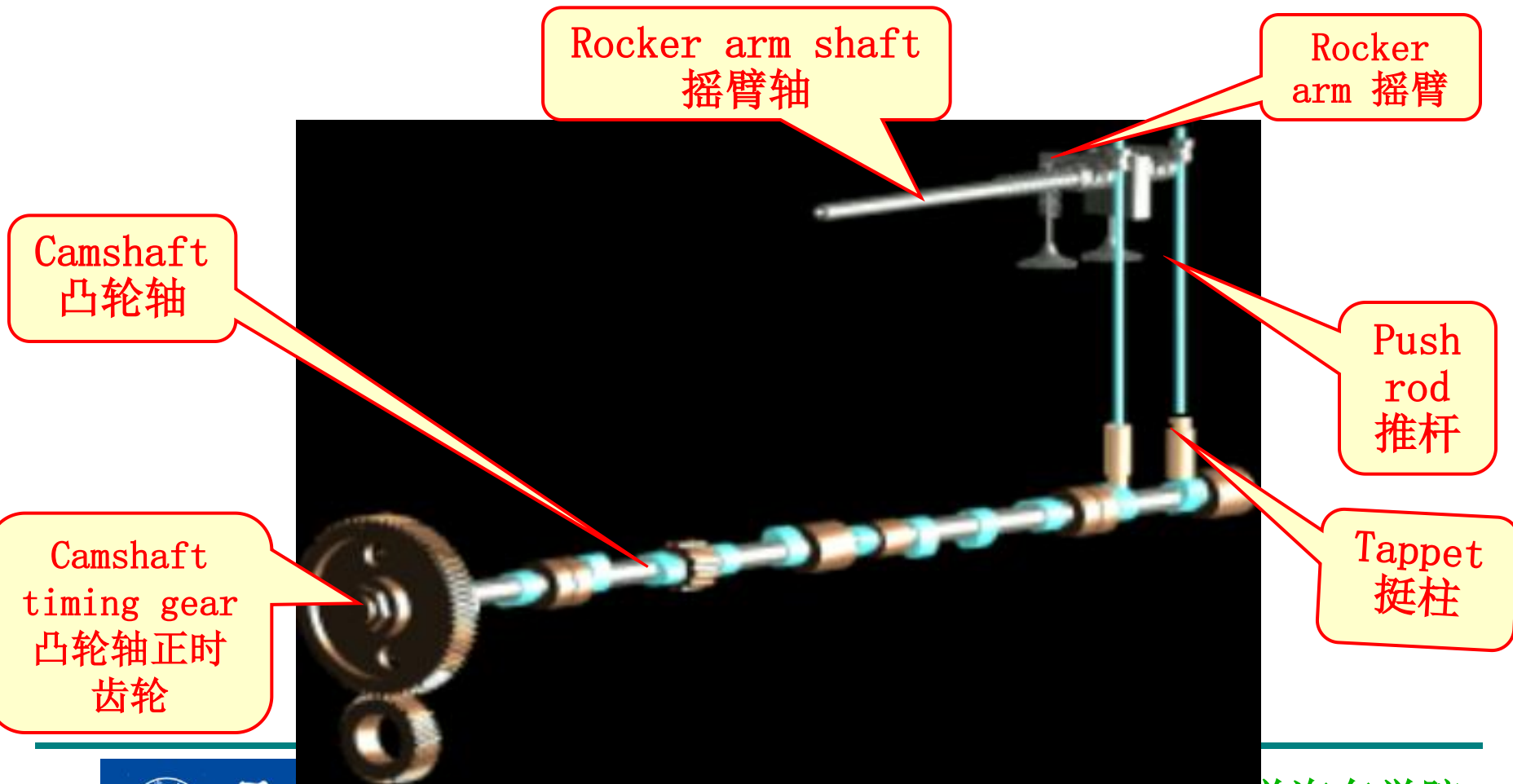


Valve Closing 气门关闭	Ensure valve closing in time and sealing 保证气门及时关闭、密封
Valve opening 气门开启	Keep valve contacting with cam 保证气门不脱离凸轮

2. Valve transmission group 气门传动组

Function: Driving valves to open and close at regular time, ensuring valves with enough opening and appropriate clearance.

功用: 定时驱动气门开闭，并保证气门有足够的开度和适当的气门间隙。



1) Camshaft 凸轮轴

Function: Drive and control valve opening and closing, making it suitable for firing order, valve timing and opening.

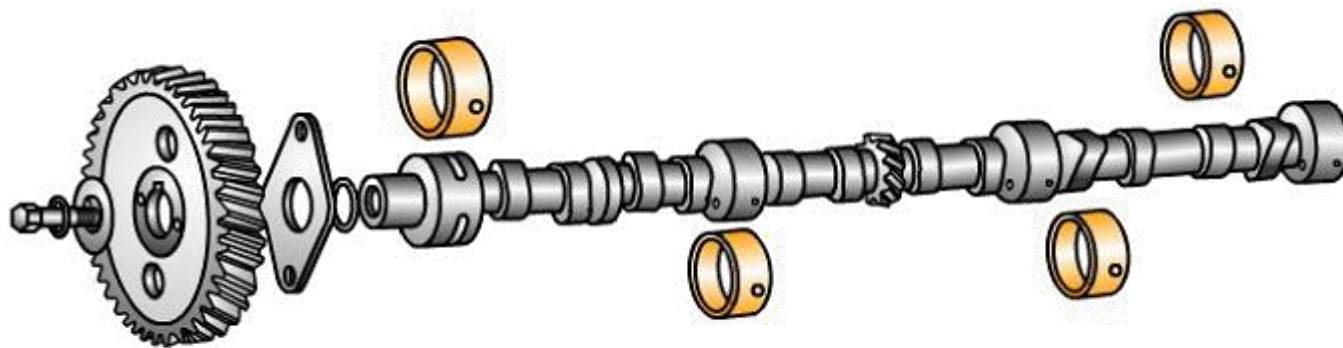
作用: 驱动和控制各缸气门的开启和关闭, 使其符合发动机的工作顺序、配气相位和气门开度的变化规律等要求。

Working condition: bear impact loads of valves intermittent open.

工作条件: 承受气门间歇性开启的冲击载荷。

Material: high-quality steel 优质钢, alloyed cast iron 合金铸铁, nodular cast iron 球墨铸铁

Structure 结构: helical gear 斜齿轮: drive distributor and oil pump 驱动分电器、机油泵; eccentric gear 偏心轮: drive fuel pump 驱动燃油泵。

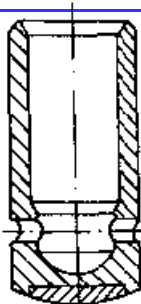


2) Tappet 挺柱

Function: Pass the thrust of cam to push rod or valve.

作用: 将凸轮的推力传给推杆或气门。

Classification 分类: plain tappet 普通挺柱 & hydraulic tappet 液力挺柱

Plain tappet 普通 挺柱	Application 用途	Graphic图示
Barrel type 筒式	Overhead valve 气门顶置式	
Roller type 滚轮式	Large bore diesel engine 多用于大缸径柴油机	

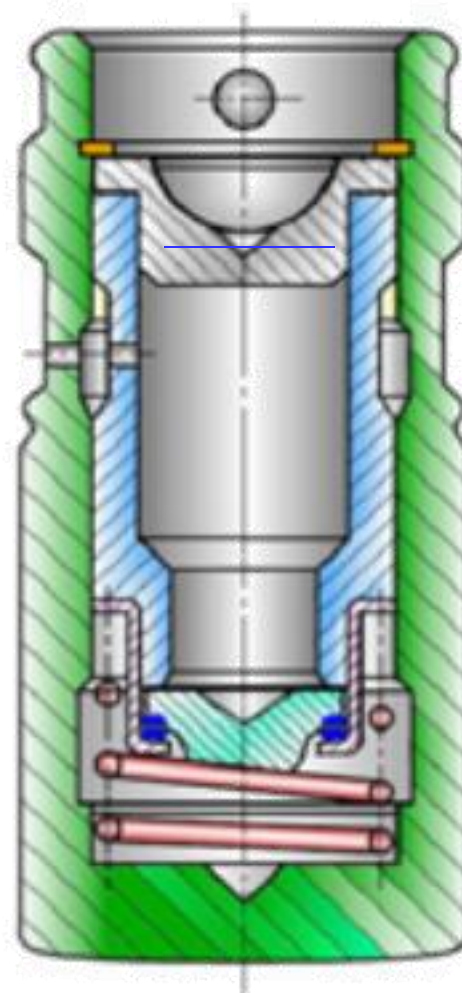


Hydraulic Tappet 液力挺柱

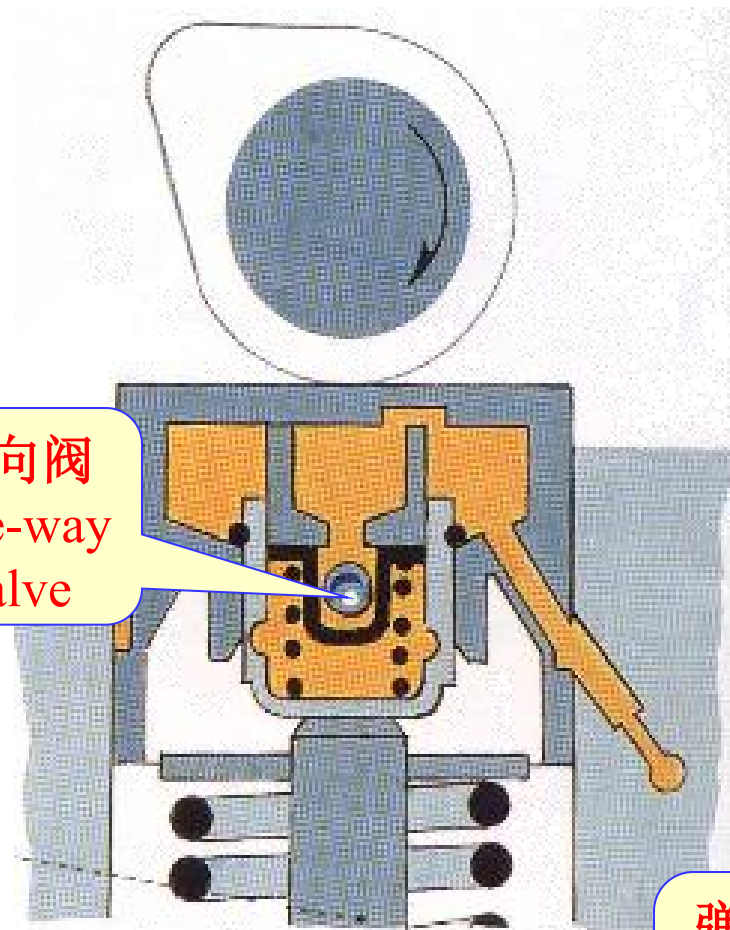
视频 (Video)

Function: Eliminate valve clearance, reduce impact loads and noise from parts, enhance high speed performance.

作用: 消除了配气机构的间隙, 减小了各零件的冲击载荷和噪声, 提高发动机高速时的性能。

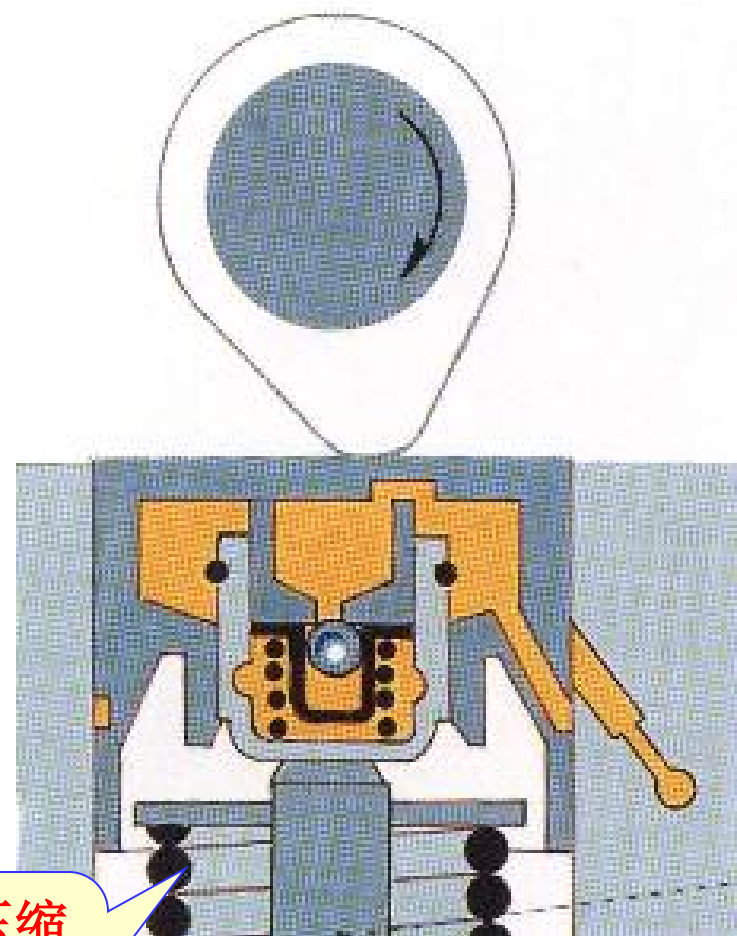


Santana Engine Hydraulic Tappet 桑塔纳发动机液力挺柱



单向阀
One-way
valve

Valve Closing



弹簧被压缩
Spring
compressed

Valve open

3) Push rod 推杆

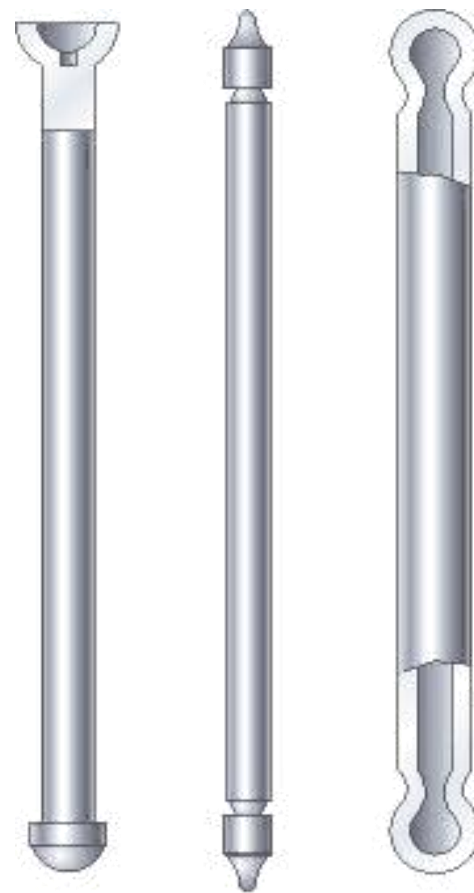
Function: Pass the thrust of tappet to rocker arm.

作用: 将挺柱传来的推力传给摇臂。

Working condition: Being the most flexible part in valve train.

工作情况: 是气门机构中最容易弯曲的零件。

Material: duralumin 硬铝 or steel 钢



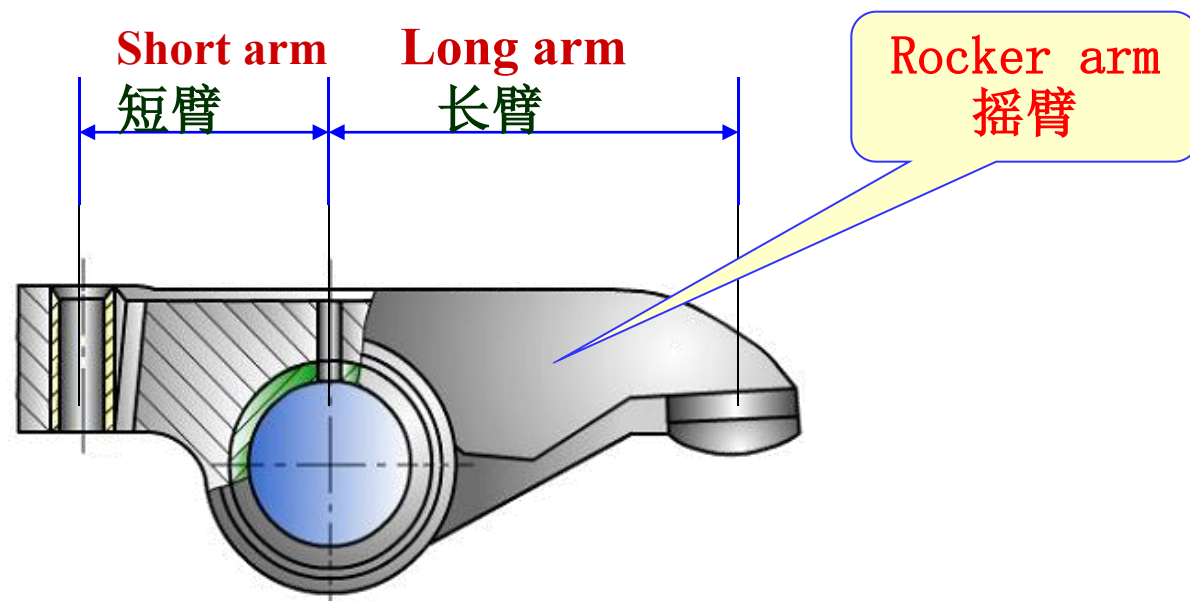
推杆

3) Rocker arm 摇臂

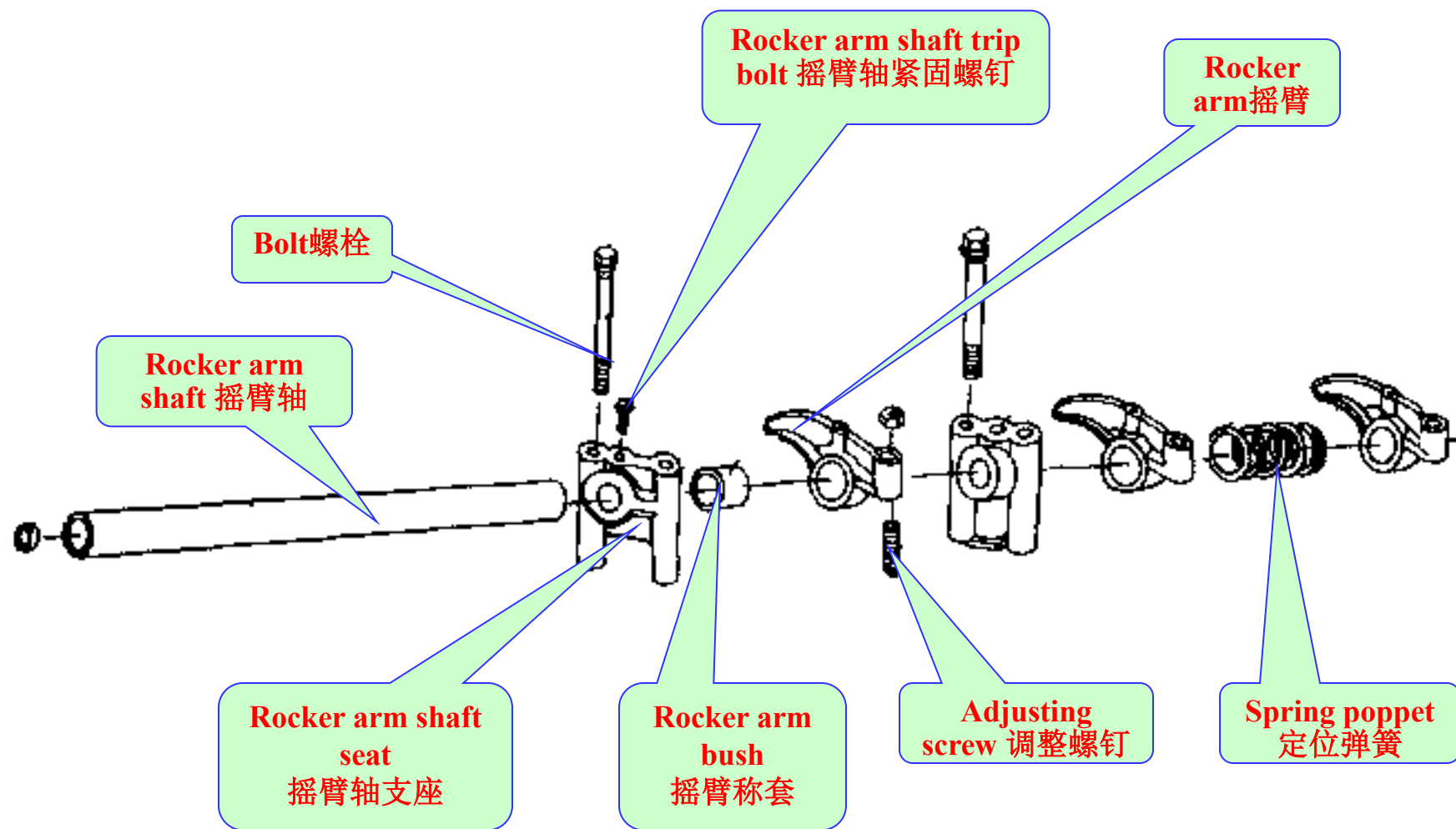
Function: change the direction of the force passed from push rod or cam to valve stem in order to open valve.

作用: 将推杆或凸轮传来的力改变方向，作用到气门杆端以推开气门。

Classification 分类: plain rocker arm 普通摇臂 & no noise rocker arm 无噪声摇臂



Rocker arm group schematic drawing摇臂组示意图



Variable Valve System(可变配气系统)



The Advantages of VVT system:

- Full range high efficiency of Volumetric Efficiency
- Higher Maximum Torque Output at low engine speed
- Higher Maximum Power Output at high engine speed
- Suitable for HCCI and new combustion control
- Suitable for HEV engine's request for engine stop without idle and to keep the three way catalyst temperature
- Other control advantages for the powertrain system

Uni-Valve System of KSPG